



### Google Allo on desktop

Google has launched a web interface for its instant messaging app. Simply scan the QR code and you're all set. <http://digit.in/gAlloWeb>



### Gran Turismo themed PS4

Sony unveils its shiny new limited edition Gran Turismo themed PS4 with a customisable controller. <http://digit.in/GranPS4>

# Origins of Modern Astronomy

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umankind has been obsessed with astronomy since its origins on Earth.

Back in ancient times, the sky was always the largest as well as the fastest moving element (apart from animals and birds, although they were pretty small compared to the sky) in the entire landscape. Over the course of history, the sky was used for a lot of purposes, ranging from keeping track of time to navigation. What we are going to do today is trace the origin of modern astronomy and figure out how we got from worshipping the sky to observing the universe like never before.

## BUT FIRST, A LESSON IN ANCIENT HISTORY

Astronomy has affected human civilisation before civilisations even existed. The earliest known records of astronomy come in the form of sticks and stones from Africa and Europe, dated as far back as 35000 BCE (used to track the moon's phases). And somewhere down that path, we landed up with a geocentric model of not only our own solar system, but an entire universe!

**Interesting fact:** Ancient Sanskrit treatise *Surya Siddhanta* asserted the spherical shape of the earth, the diameter as 8000 miles (modern: 7,928 miles), the moon's diameter as 2,400 miles (actual ~2,160) and the distance between moon and earth to be 258,000 miles (actual ~238,000) as early as 4th century CE.

We consistently believed, up until the late 16th century, that the Earth was the centre of the universe. This could be chalked up to the lack of proper measuring instruments or techniques. It wasn't until the helio-

Gazing at the sky has become a lot clearer – and important – over the years.

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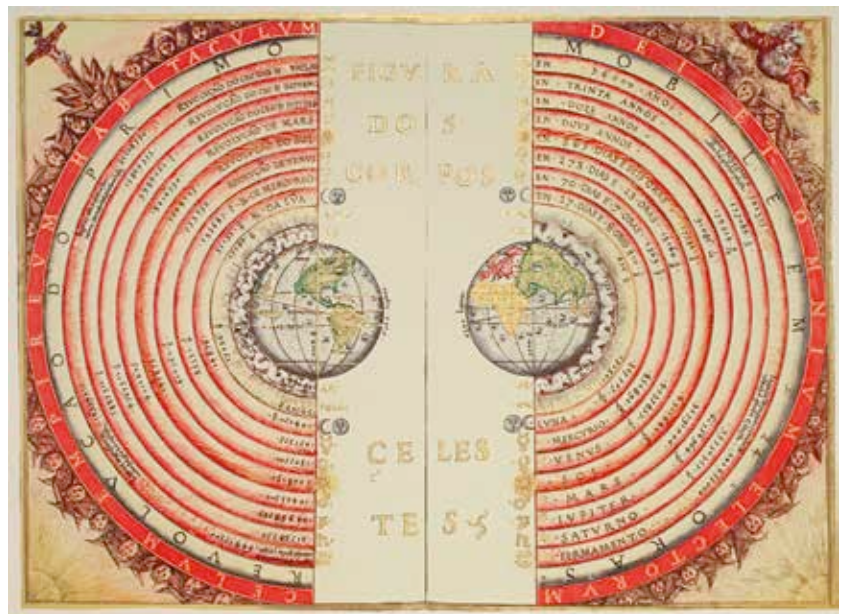
centric model was proposed by Copernicus that the earlier geocentric theory was shunned and replaced.

## ASTRONOMY POST GALILEO AND THE TELESCOPE

Around the beginning of the 17th century, a new type of observational device was being built and popularised. The refracting telescope was initially invented by Dutch eyeglass maker Hans Lippershey and was going to be one of the most significant inventions in the history of astronomy. With the

quick proliferation of his design and other inspired alternatives, astronomy saw a revolution like never before.

Within the next two years, two of the most significant contributors to modern astronomy, Johannes Kepler and Galileo Galilei, published their seminal works. While Kepler's laws of planetary motion shunned the circular orbits of planets for elliptical ones, Galileo recorded observations with his improvised telescope that had never been seen before – spots on the sun, craters on the moon and more.



The geocentric model